A Multi-Agent LLM Approach to Virtual Museum Exploration

Group 49

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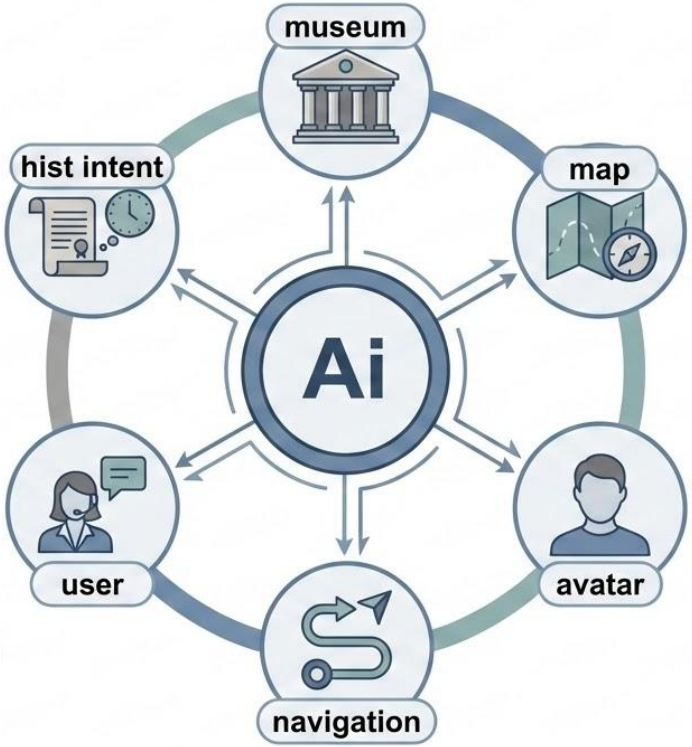
Problem Statement

We have environment and we used AI to make it interesting and interactive

Current Solutions:

- Limited interactivity
- Restriction to following pre-set tour
- Reading pre-set text
- “One-size-fits all”

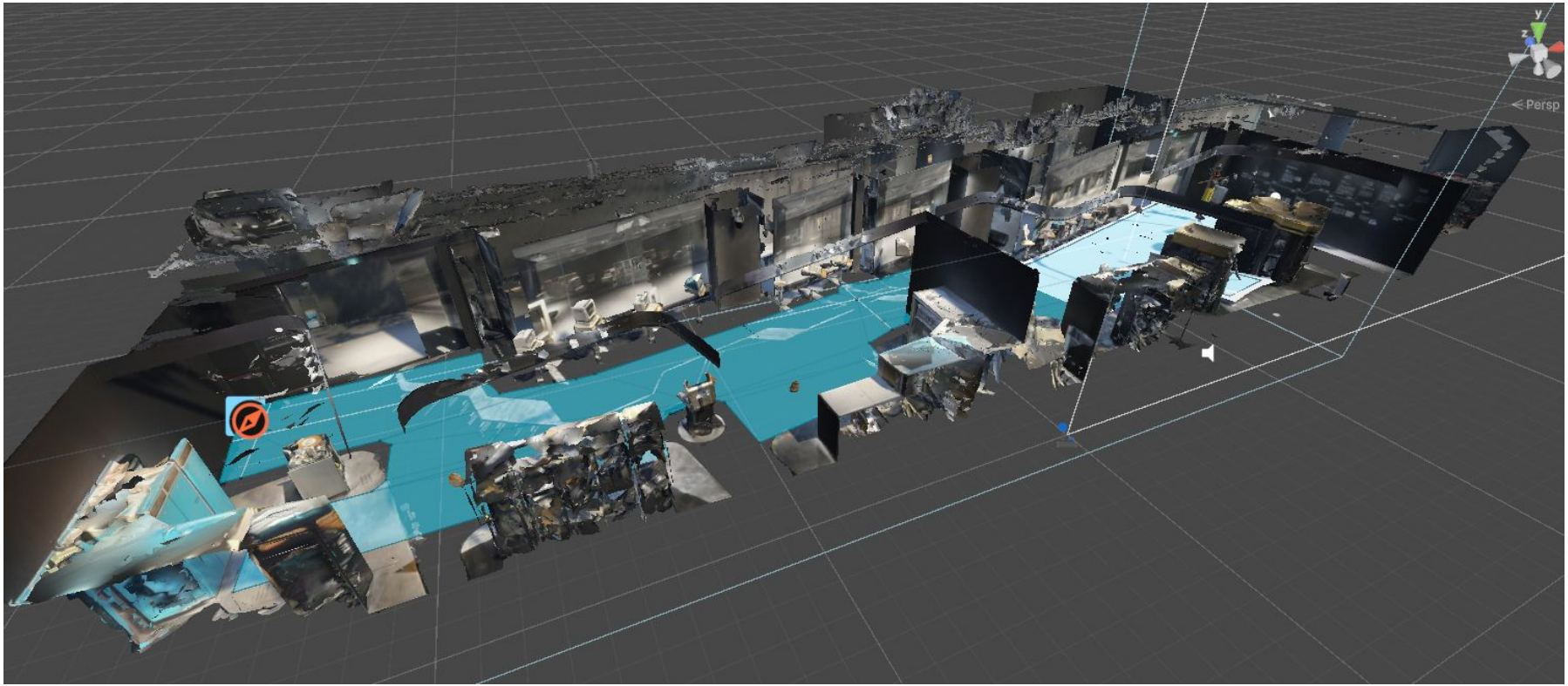


Solution and Objectives

We propose a **Dynamic Conversational Tour Guide** driven by AI.

- **Context-Aware:** The AI knows where you are and what you have seen.
- **Personalized:** Tours adapt to the user's interests.
- **Multi-Modal:** Supports both voice and text interaction.

Setup - Virtual Museum in Unity



Introducing Interactive Avatar Guide

The Avatar is a bridge between the **user** and the **environment**.

User to Avatar:

- Users talk to the Avatar using **voice** or by **typing** in an input field.

Avatar to User:

- The Avatar answers with **voice** or shows **feedback** on the screen.

Avatar to Environment:

- The Avatar moves and acts in the environment using NavMesh.

Design - Unity and Gemini - Proxy Python Server

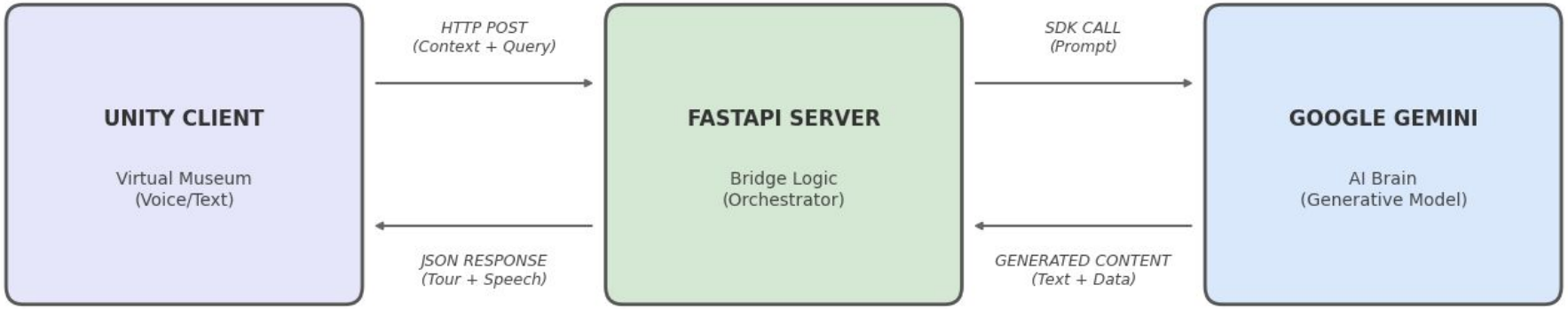
1. **Unity Client**
   - Gets voice input.
   - Sends data to FastAPI

2. **Python FastAPI (Bridge)**
   - FastAPI runs logic Unity can't
   - Gets data, finds intent, calls Gemini

3. **Google Gemini (The Brain)**
   - Processes instructions and context
   - Sends JSON to Unity Avatar

```
Response = model.generate_content(...)

   # Return properly structured response to unity
    return {
        "content": content_string,  # Now it's
always a string
        "tasks": [task]
    }
```



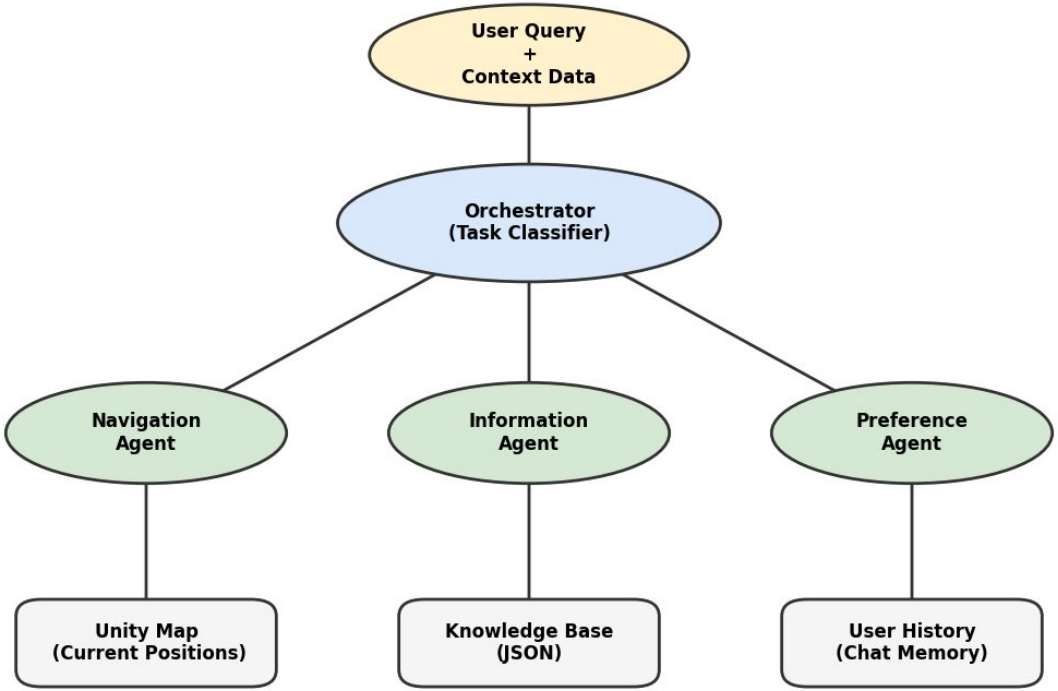
AI - Multi-Agents Orchestration

Agents

- Classifier Agent
- Navigation Agent
- Information Agent
- Preference Agent

Tools

- History
- Map
- Knowledge Base



Agent 0 - Classifier Prompt

```
def get_classify_prompt(query: str, current_object: str, current_position: str, visited_objects: str, museum_exhibits: str) -> str:
    return f"""I have a classification task about interactive types in the virtual "Beyond Computing Museum" (台灣電腦資訊發展館)
with four labels: "information", "navigation", "preference", and "error".
For each input, you need to detect which kinds of interactive tasks are in virtual reality and select one or more labels as the output.

"information" refers to questions that require information to enhance the user's understanding of the virtual environment or a specific exhibit.
Or the input is similar to "please explain more."

"preference" refers to descriptions related to users themselves, like their background and personal interests,
which could be used to tailor the tour.

"navigation" means you want to move or go to another exhibit, or visit the whole virtual environment.
You might say: "take me somewhere" or "give me a tour." or "show me those exhibits.", "yes, I would like to go with this tour", "okay i would like to go with this tour"
"next" is also a navigation intent. CONFIRMS/ACCEPTS starting a tour (e.g., "I would like to start this tour", "Yes", "Let's go") is also a navigation intent.
If the input is not related to the museum's exhibits, it will be defined as the "error" label.

Here are examples:

INPUT: Please give me a tour to visit this virtual museum.
RESPONSE: information

INPUT: I want to see Hard Disk Drive.
RESPONSE: navigation

INPUT: I am a computer science professor.
RESPONSE: preference

INPUT: Next
RESPONSE: navigation

INPUT: I want to see some minicomputers
RESPONSE: navigation, information, preference
```

Agent 1 - Navigation

Navigate user to the exhibit

Details:

- Only valid JSON.
- Avoids go to the same place twice.
- Dual-channel response

```
INPUT: What are the three most popular items in this museum
RESPONSE:
As a helpful tour guide, I can provide you with information about the three most popular exh
To provide you with a general idea, some of the most well-known exhibits in this museum incl
and "CDC Cyber 170/720 Mainframe computer."
These exhibits are widely recognized and admired for their historical significance in comput
However, it's important to note that popularity can change over time and may differ based on
INPUT: Take me to visit them.
RESPONSE:
{{
    "introduction": "Now let's begin our tour. First, we will head towards the 'PDP-10 Mainf
    and revolutionized computing with its time-sharing system.",
    "tour": ["PDP-10 Mainframe Computer", "Apple II Personal Computer", "CDC Cyber 170/720 M
    "tour_ids": ["exhibit_026", "exhibit_016", "exhibit_019"]
}}
INPUT: I have visited these exhibits ["exhibit_026"]. I want to see the next exhibit.
RESPONSE:
{{
    "introduction": "Now, let's move on to the 'Apple II Personal Computer'. Follow me as we
    "tour": ["Apple II Personal Computer", "CDC Cyber 170/720 Mainframe computer"],
    "tour_ids": ["exhibit_016", "exhibit_019"]
}}
```

Agent 2 - Information

Tell user about exhibit

Features:

- Uses a JSON knowledge base
- Knows where user is
- Doesn’t repeat exhibits you’ve seen

JSON database


```
"exhibits": [
  {
    "id": "exhibit_001",
    "title": "Flash Memory",
    "artist": "Fujio Masuoka (Toshiba)",
    "year": "1984",
    "description": "A non-volatile memory that can retain data even without power.",
    "details": "Flash memory offers better resistance to shocks compared to traditional hard drives and is widely used in mobile devices
    "style": "Storage"
  },
  {
    "id": "exhibit_002",
    "title": "Hard Disk Drive",
    "artist": "IBM",
    "year": "1973",
    "description": "A compact, high-capacity auxiliary storage device. The modern HDD architecture was established by the IBM 3340.",
    "details": "The company that developed the first HDD for desktop computers is now known as Seagate.",
    "style": "Storage"
  },
```


```
Here are some examples:

CURRENT EXHIBIT: Apple II Personal Computer
USER QUERY: Tell me about this.
RESPONSE: {{"response": "The Apple II personal computer was introduced by Apple Inc. in 1977. It was the world's first widely popular personal computer s

CURRENT EXHIBIT: CDC Cyber 170/720 Mainframe computer
USER QUERY: What is this exhibit?
RESPONSE: {{"response": "This is the CDC Cyber 170/720 mainframe computer from 1975. It was used at NCTU to process the University Joint Entrance Examina

USER QUERY: Are there any exhibits that were used at NCTU?
RESPONSE: {{"response": "Yes, there are several exhibits in the museum that were used at National Chiao Tung University (NCTU). Here are some of them:\\n

CURRENT EXHIBIT: IBM PC-XT Personal Computer
USER QUERY: I want to finish the tour now.
RESPONSE: {{"response": "Thank you for visiting the Beyond Computing Museum! We hope you enjoyed learning about the history of computing. Feel free to ex

Museum exhibits information:
{museum_exhibits}
```

Agent 3 - Preferences

Captures user background

Influences future Navigation and Information responses:

- Student get a different tour suggestion than a professor

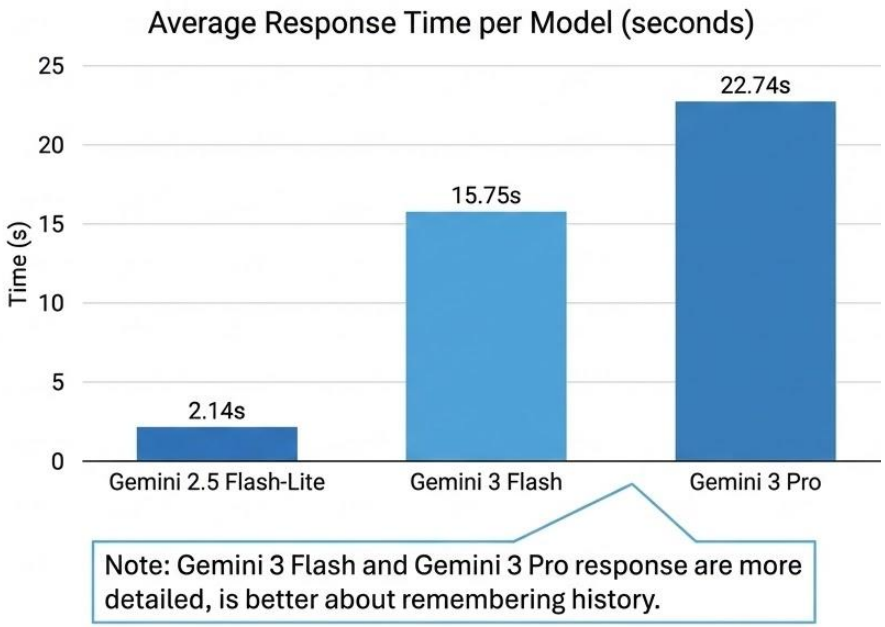
```
def get_preference_prompt(query: str, current_object: str, current_po
    return f"""You are a helpful tour guide for the Beyond Computing

Acknowledge their preferences and provide a brief, friendly response
If appropriate, mention how this might help personalize their museum

Respond with only a JSON object containing a 'response' field (no mar
```

Which Gemini?

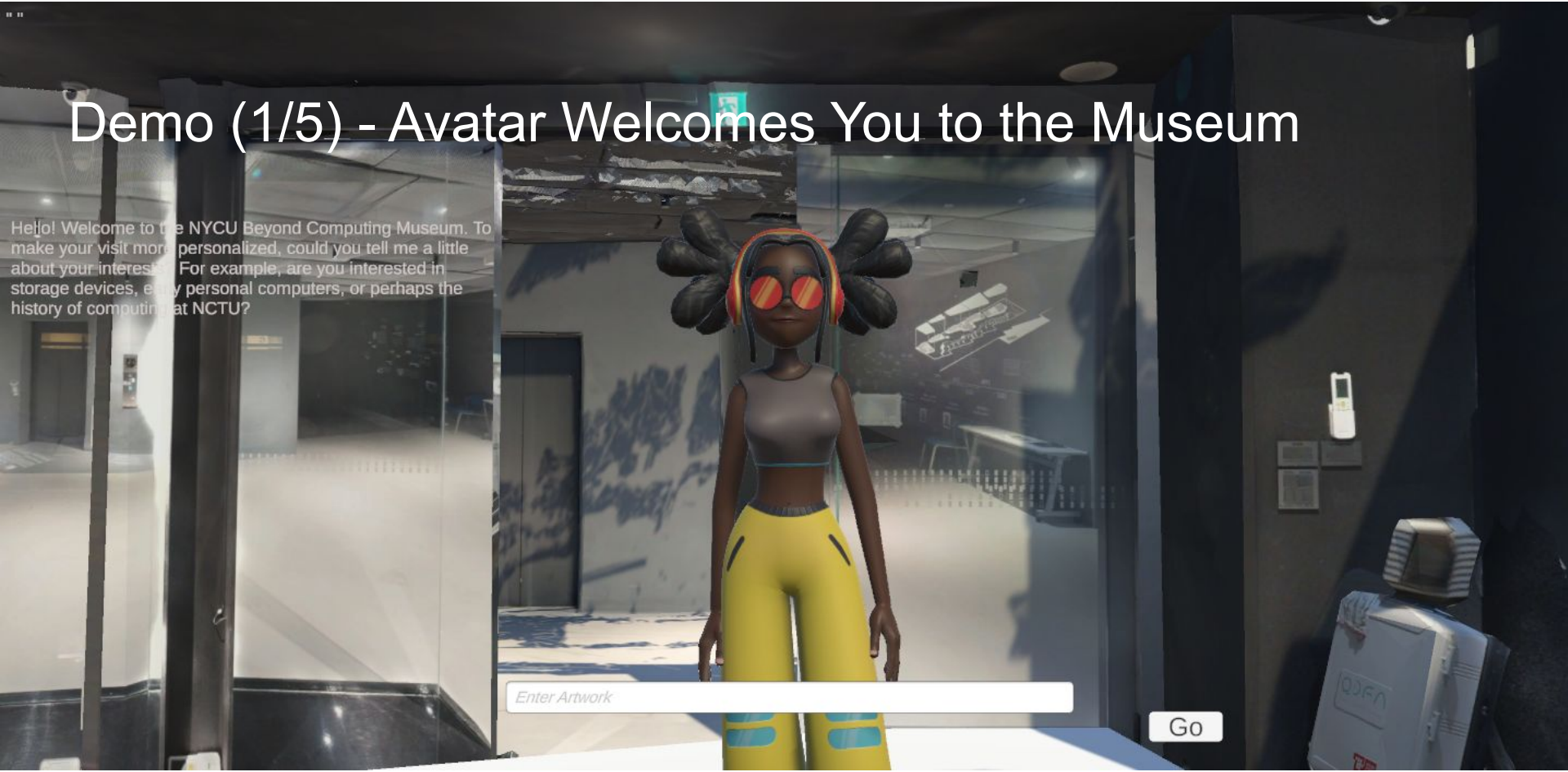
We prioritise latency so choose Gemini 2.5 Flash-Lite



Latency Trade Off

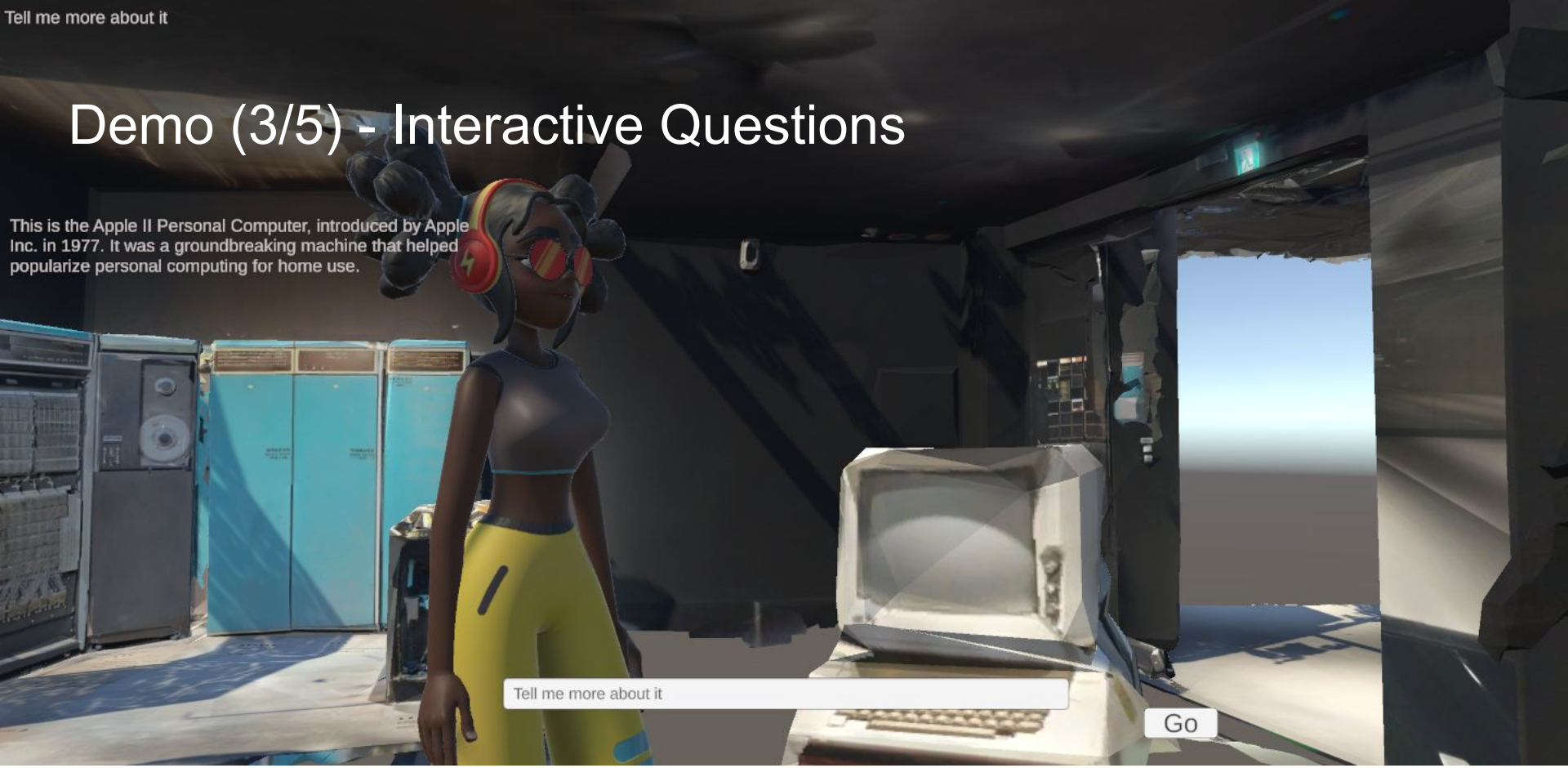
- 1 is faster
- 2, 3 is more detailed

Demo (1/5) - Avatar Welcomes You to the Museum

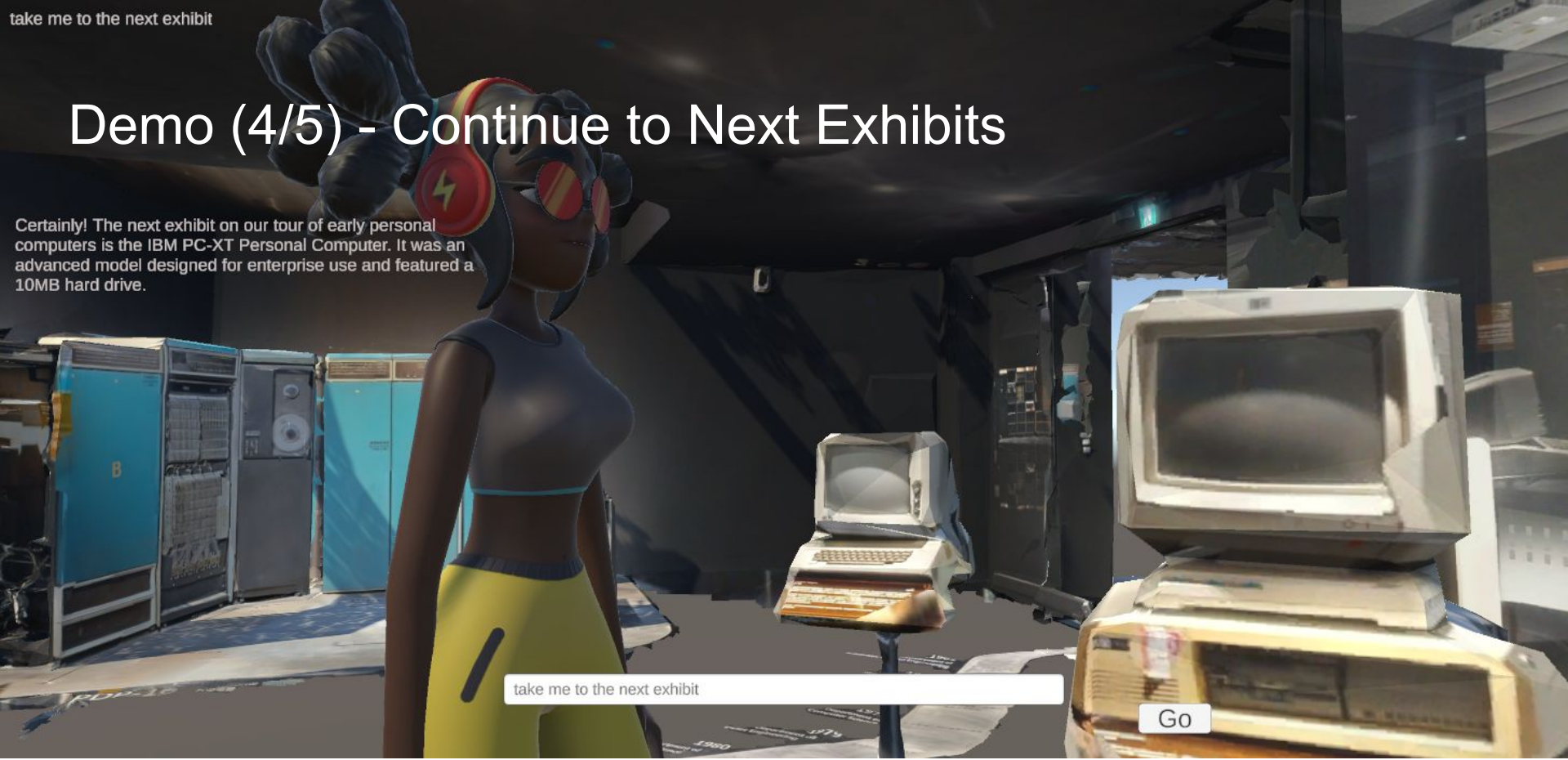


Demo (2/5) - First Prompt

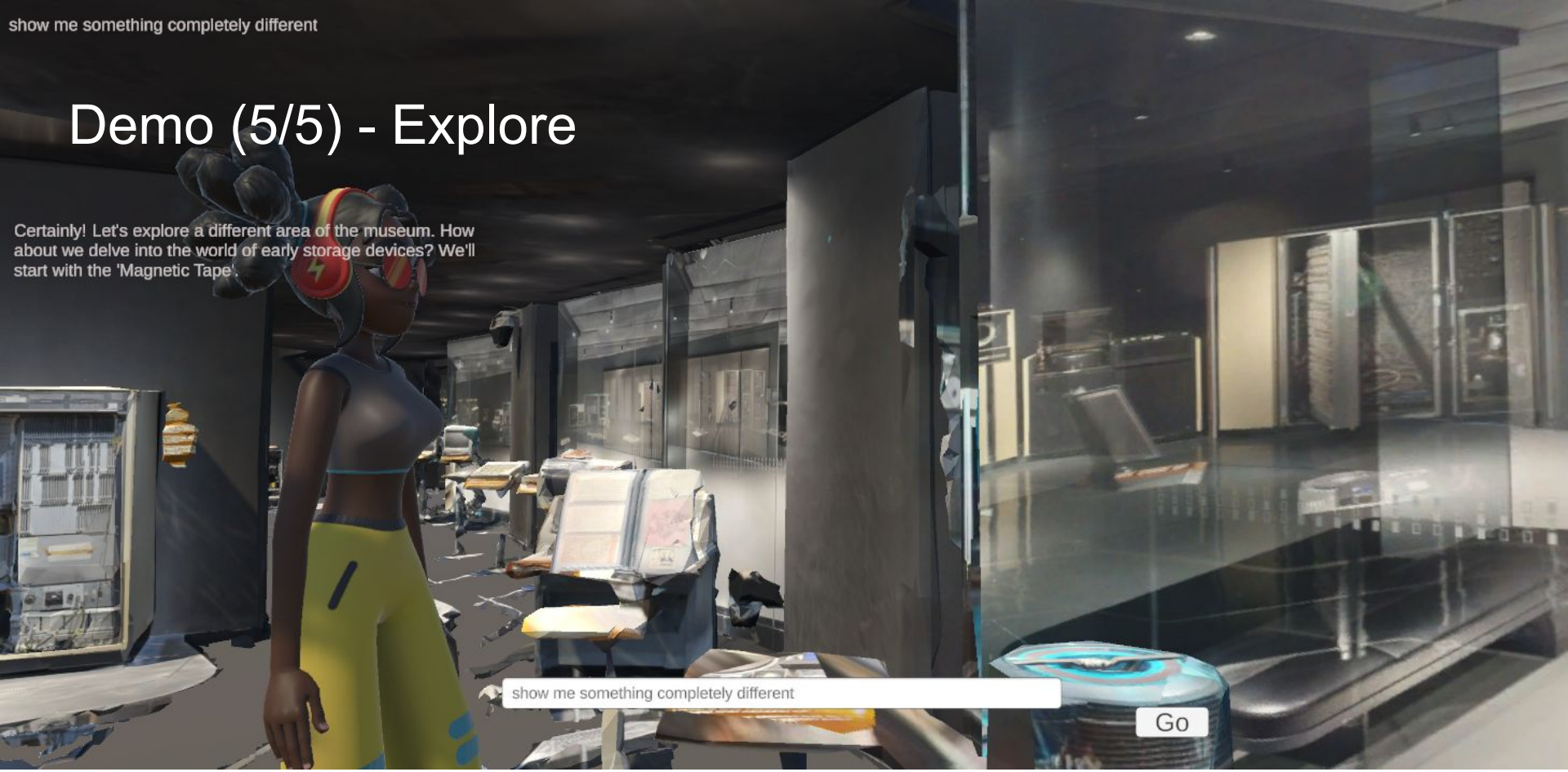
Demo (3/5) - Interactive Questions



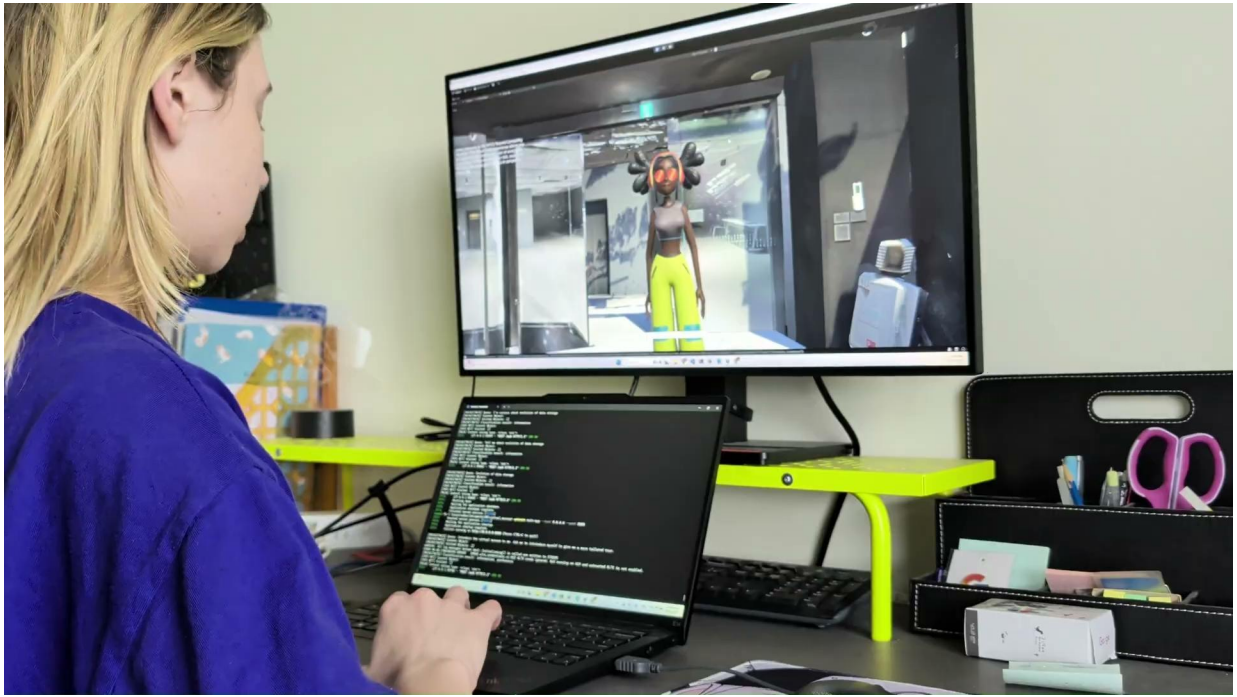
Demo (4/5) - Continue to Next Exhibits



Demo (5/5) - Explore



Demo Video



Conclusion

We implemented a LLM driven avatar to enhance the museum experience.

- **Successful Integration:** Bridged Unity and Google Gemini via a Python proxy server.
- **Context-Awareness:** The AI is given the user context and a memory component to provide relevant information.
- **Dynamic Personalization:** Moved beyond static, linear tours to dynamic paths tailored to user interests (e.g., "Show me mainframes").
- **Real-Time Performance:** Optimized latency (~2s) using Gemini 2.5 Flash-Lite to maintain immersive interaction.

Future Work

- ADK - Agent Development Kit
- Gemini 3 Models: Flash -> Pro
- Increased Agent Capabilities and Multi-Agent Control Plane (MCP)